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**4** **(a)** **(i)** State what is meant by the internal energy of a system.

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..... [2]

**(ii)** Explain why the internal energy of an ideal gas is directly proportional to the thermodynamic temperature of the gas.

.....

.....

.....

..... [2]

**(b)** A sample of an ideal gas at thermodynamic temperature $T$ has internal energy $U$.

The gas is compressed so that its temperature increases to $3T$.
During this compression, work $W$ is done on the gas.

The gas is then cooled at constant volume so that its temperature decreases to $2T$.

Complete Table 4.1 to show, in terms of some or all of $W$, $T$ and $U$, the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas for each of the two processes.

**Table 4.1**

| | work done on gas | thermal energy supplied to gas | increase in internal energy of gas |
|---|---|---|---|
| compression | $+W$ | | |
| cooling | | | |

[4]

[Total: 8]